

V1: Forward-Model Mismatch

4 October 2026

What parameter bias does forward-model error cause?

Amir Dollie (DLLAMI004), EEE4022S

SHARC buoy project Code: `runV1ModelMismatch.m`, `v1Tools.m`, `diagnoseV1Truncation.m`

Results: `Results/V1/V1_20261004_121316` (commit 1711fb3)

Summary

Question. Every study so far generated and inverted with the same EMM (the “inverse crime”); M1 showed that the estimator behaves as predicted when the model is right. V1 generates the data from a *perturbed* truth model and keeps the frozen nominal inverse, to separate forward-model error from estimator uncertainty. **Outcome.** The three mismatch sources behave completely differently. **Truncation** (production [50 10 10] against the converged [130 20 20]): $d_{\text{sys}} = 0.77$ to 3.22 , which moves p_0 from Identifiable to Marginal and two large- kR floes from Marginal to Not identifiable, although the parameter shifts are below 0.4% in β and 0.13% in R . **Water depth** $\pm 5\%$: benign, $d_{\text{sys}} \leq 0.205$ and $|\delta\beta/\beta| \leq 0.048\%$; no tolerance is crossed, so the depth tolerance exceeds $\pm 5\%$ over the tested range. **Common radial sensor error:** dominant. $d_{\text{sys}} = 0.5$ at $|\Delta r| \approx 0.050$ mm, $d_{\text{sys}} = 1$ at 0.10 mm and $|\delta\beta/\beta| = 1\%$ at 0.29 mm; centimetre errors are far outside the linear regime (confirmed by a nonlinear inversion). Sensor radial coordinates relative to the model floe geometry must be known or jointly estimated to sub-millimetre precision.

1 What the study does

Why V1 exists. A real floe is never exactly the EMM: the series is truncated, the depth is measured with error and the sensors are not exactly where the model places them. With $\sigma_{\ln \beta} = 0.36\%$ and $\sigma_{\ln R} = 0.067\%$ at p_0 , a sub-percent model-induced shift can be statistically decisive; V1 measures it in units of that uncertainty.

Method (pre-registered in `cfg.V1`; clarifications and one pre-run correction logged before the run).

- Systematic bias.** $b_{\text{sys}} = f_W(\text{truth}) - f_W(\text{nominal})$, both the Welch-expected features from one function, projected with the *nominal* J , Σ and F : $\delta\theta_{\text{sys}} = F^{-1}J^T\Sigma^{-1}b_{\text{sys}}$, $d_{\text{sys}} = \sqrt{\delta\theta_{\text{sys}}^T F \delta\theta_{\text{sys}}}$. The class uses d_{tot} from the *vector* sum $\delta\theta_W + \delta\theta_{\text{sys}}$.
- V1a truncation.** Stage 0 selected the truth truncation by a ladder at p_0 ($M = 90, 110, 130, \dots$ with $[PN] = [20\ 20]$) until a step moved the features by $d < 0.1$. Inverse [50 10 10] against that truth at p_0 and the three large- kR A1 floes.
- V1b water depth.** True $H = H_0(1 + \epsilon_H)$, $\epsilon_H \in \{\pm 2, \pm 5\}\%$, physically consistent ($\beta \propto H^{-4}$; γ , R , $r_i \propto H^{-1}$), physical sea and frequency band fixed.
- V1c radial sensor position.** Every sensor ($r > 0$) moved by a common signed $\Delta r \in \{\pm 10, \pm 20\}$ mm, angles unchanged.
- Tolerances.** First crossing per side of $d_{\text{sys}} = 0.5, 1$ and $|\delta\beta/\beta| = 1\%$, bracketed by the registered points and bisected in $\log_{10} |\text{error}|$ to 0.005 decade (pre-run correction: an absolute tolerance could not resolve crossings near 0.05 mm). Not extrapolated beyond the registered range.
- Nonlinear check** (diagnostic). The largest- d_{sys} case inverted with G3’s solver from θ_{true} on its noiseless Welch features, compared with $\theta_{\text{true}} + \delta\theta_W + \delta\theta_{\text{sys}}$.

Depth and radial cases are at p_0 (Level 2C, s26, LSM6DSV16X, setting F). The nominal Welch features reproduce the engine’s $f_0 + b_W$ exactly, so b_{sys} has no implementation offset. Clean tree enforced (commit 1711fb3, 59 min).

2 Results

2.1 Stage 0 and V1a: truncation

The p_0 ladder moved by $d = 0.125$ from $M = 90$ to 110 and by $d = 0.079$ from 110 to 130 , so [130 20 20] was selected. The same last step at the three large- kR floes gave $d = 0.062, 0.057$ and 0.069 , and raising P to 30 at $M = 130$ gave $d = 0.050$: the selected truth is converged by the registered rule at every V1a floe and in P .

Floe	d_{sys}	$\delta\beta/\beta$	$\delta R/R$	d_W	d_{tot}	Class: matched $\rightarrow$ mismatched
p_0	1.024	-0.031%	$+0.032\%$	0.309	0.905	Identifiable $\rightarrow$ Marginal
$R \times 2$	1.831	$+0.163\%$	$+0.063\%$	0.617	1.547	Marginal $\rightarrow$ Not identifiable
$\beta \times 0.5, R \times 2$	3.220	$+0.383\%$	$+0.124\%$	0.593	3.066	Marginal $\rightarrow$ Not identifiable
$\beta \times 2, R \times 2$	0.771	$+0.065\%$	$+0.024\%$	0.587	0.631	Marginal $\rightarrow$ Marginal

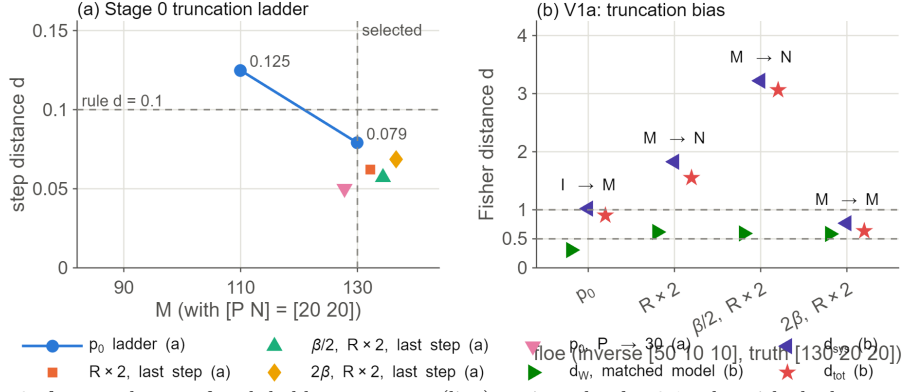


Figure 1. (a) Stage 0: feature change of each ladder step at p_0 (line) against the $d = 0.1$ rule, with the last step $[110 \rightarrow 130]$ at the large- kR floes and the $P \rightarrow 30$ check (markers offset slightly in M). (b) V1a per floe: the Welch bias of the matched model d_W , the truncation bias d_{sys} and their vector total d_{tot} , with the class change (I, M, N); ticks at $d = 0.5$ and 1 are the class limits.

2.2 V1b and V1c: depth and radial position

Case	d_{sys}	$\delta\beta/\beta$	$\delta R/R$	d_{tot}	Class
$\epsilon_H = -5\%$	0.205	-0.048%	-0.002%	0.144	Identifiable
$\epsilon_H = -2\%$	0.077	-0.017%	-0.001%	0.242	Identifiable
$\epsilon_H = +2\%$	0.071	+0.015%	+0.000%	0.373	Identifiable
$\epsilon_H = +5\%$	0.169	+0.033%	+0.000%	0.462	Identifiable
$\Delta r = -20$ mm	208.6	+106.9%	+13.9%	208.8	Not identifiable
$\Delta r = -10$ mm	102.3	+42.6%	+6.7%	102.6	Not identifiable
$\Delta r = +10$ mm	98.4	-28.6%	-6.1%	98.1	Not identifiable
$\Delta r = +20$ mm	192.8	-48.2%	-11.7%	192.6	Not identifiable

Radial parameter biases are the linear projection; see the nonlinear check before reading them literally.

Tolerances. Depth: none of the six (two sides, three criteria) is crossed within $\pm 5\%$ (largest $d_{\text{sys}} = 0.205$, largest $|\delta\beta/\beta| = 0.048\%$). Radial, first crossing per side (negative / positive): $d_{\text{sys}} = 0.5$ at ∓ 0.0498 mm, $d_{\text{sys}} = 1$ at ∓ 0.0996 mm, and $|\delta\beta/\beta| = 1\%$ at $-0.288 / +0.290$ mm.

The α range of the depth cases. Rescaling H rescales α . The *retained* bins stay inside the validated window $[1.7, 13.85]$ in all four cases (at $+5\%$ the retained maximum is 13.807), but the wider synthesis support leaves it (support $[1.615, 13.157]$ at -5% , maximum 14.542 at $+5\%$). The runner flags all four as extrapolated; this is a caveat on the support region, not a failure of the retained features.

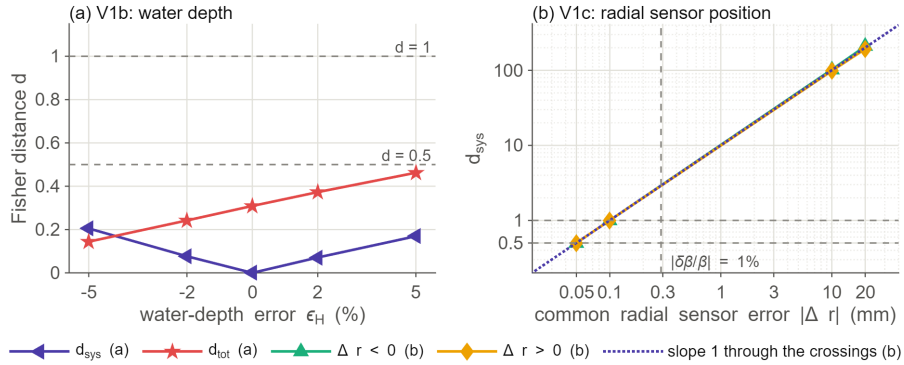


Figure 2. (a) d_{sys} and d_{tot} against the water-depth error (the $\epsilon_H = 0$ point is the matched model: $d_{\text{sys}} = 0$, $d_{\text{tot}} = d_W$). (b) d_{sys} against the common radial sensor error, both signs, log-log: the registered ± 10 and ± 20 mm points and the bisected crossings of $d = 0.5$ and 1 (ticks); dotted: slope 1 through the crossings; dashed vertical: the $|\delta\beta/\beta| = 1\%$ crossing.

2.3 Nonlinear check

The largest d_{sys} is $\Delta r = -20$ mm. The solver converged in 8 iterations (36 model evaluations, 20 min) to $\delta\beta/\beta = +80.5\%$ and $\delta R/R = +12.6\%$, against the linear $+107.1\%$ and $+13.9\%$. The gap to the linear total is $[-38.7, -17.3]\sigma$ ($d = 47.7$; the 0.2σ reference is not a gate for a diagnostic this far out).

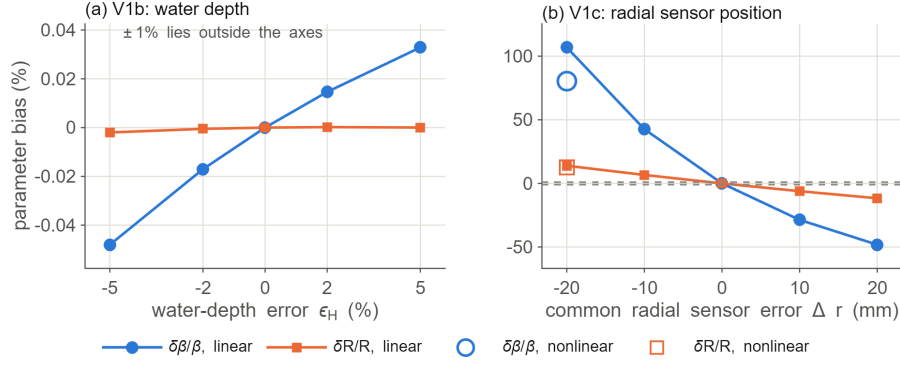


Figure 3. Linearised parameter bias against (a) depth error and (b) radial error, with the nonlinear inversion at $\Delta r = -20$ mm (open markers). In (b) the dashed lines at $\pm 1\%$ sit almost on zero; in (a) they lie outside the axes.

3 Interpretation

1. **Converged metrics, but not a converged model.** A1 concluded that σ and the class had converged in the truncation, and that remains true. V1a shows that the *features* of [50 10 10] still differ from the converged model by more than the estimator can tolerate: the parameter shifts are sub-percent, yet because the inverse is so precise they amount to $d_{\text{sys}} = 0.77$ to 3.22. The model is not grossly wrong physically; its discrepancy is statistically significant. For real data the inverse should use [130 20 20] (about 100 s per field against about 30 to 40 s).
2. **Depth is benign.** All four cases stay Identifiable and no registered threshold is crossed: the depth tolerance exceeds $\pm 5\%$ over the tested range. No numerical tolerance is extrapolated beyond it.
3. **Radial geometry is the dominant practical model-validity problem.** d_{sys} grows linearly with $|\Delta r|$ (slope 1 in Figure 2b, about 10 per mm) and is almost exactly symmetric in sign at the crossings (± 0.0498 and ± 0.0996 mm), which marks it as a genuine small-perturbation result rather than an artefact of the far-field linear extrapolation. For scale, $\sigma_{\ln R} = 0.067\%$ is 0.48 mm on the basin floe radius, so a common sensor shift of about one tenth of the radius uncertainty already biases the estimate by half a standard deviation.
4. **The centimetre-scale numbers are qualitative.** Given the 47.7 Fisher-unit nonlinear gap, the ± 10 and ± 20 mm parameter biases must not be read literally; they show only that such errors lie far outside the local regime. The quantitative radial result is the set of small-error crossings at 0.05 to 0.29 mm.
5. **Biases add as vectors.** At p_0 the truncation and Welch biases partly cancel ($d_{\text{sys}} = 1.024$, $d_W = 0.309$, $d_{\text{tot}} = 0.905$); at $+5\%$ depth they partly reinforce ($d_{\text{sys}} = 0.169$, $d_{\text{tot}} = 0.462$), and at -5% they cancel (0.205, 0.144). Adding distances instead would put p_0 at 1.33 (Not identifiable rather than Marginal) and the $\beta \times 2$, $R \times 2$ floe at 1.36 rather than 0.63, which justifies the vector combination fixed in the method.

Conclusion. Breaking the inverse crime exposed forward-model error as a distinct limitation from estimator uncertainty. Increasing the EMM truncation from [50, 10, 10] to the converged [130, 20, 20] produced Fisher-scaled systematic biases of $d_{\text{sys}} = 0.77$ to 3.22 and changed several identifiability classifications, despite sub-percent parameter shifts. Water-depth errors of $\pm 5\%$ were benign over the tested range, with $d_{\text{sys}} \leq 0.205$. In contrast, coherent radial sensor-coordinate errors were extremely influential: $d_{\text{sys}} = 0.5$ and 1 occurred at approximately 0.05 and 0.10 mm respectively, and a 1% stiffness bias occurred near 0.29 mm. Large centimetre-scale errors lay far outside the linear regime, as confirmed by the nonlinear inversion. Consequently, geometric uncertainty must be calibrated or incorporated explicitly into the inverse rather than treated as negligible.

Limitations. V1c tests one coherent radial shift common to all sensors, not independent random, angular or GPS-type errors, which may partly average out or be absorbed by R . Biases are linear projections, checked nonlinearly only at the extreme radial case (an optional solve near $+0.1$ mm would confirm the $d \approx 1$ tolerance directly). Depth and radial cases are at p_0 only; truncation covers four floes. Beyond the deliberate mismatch, truth and inverse share the EMM physics, so other physical model error is outside V1. The [130 20 20] truth is converged to the $d < 0.1$ step rule, not exactly.

Requirement for the experiment and the thesis

Truncation: invert real data with [130 20 20] (or re-run the stage 0 ladder for the floe at hand); [50 10 10] remains adequate for the σ -based design and domain trends (A1 showed σ converged), but the bias-based classes of those studies are within-model and exclude truncation bias. **Depth:** measurement to within $\pm 5\%$ suffices. **Geometry:** the radial coordinates of the sensors relative to the assumed floe centre and model geometry must be known or jointly estimated to sub-millimetre precision; centimetre-scale geometric uncertainty cannot be treated as negligible. Remedies: geometric calibration, photogrammetry, or adding position parameters to the inverse (to be planned with E0). The Conclusion above is the thesis wording.

Outputs (Results/V1/): V1_20261004_121316.log/.mat/_summary.txt; V1_truncation.csv, V1_mismatch.csv, V1_tolerances.csv, V1_nonlinear.csv; diagnostics/truncationLadder_20261004_113129.log/.mat; figures/V1_fig1 to V1_fig3 (.pdf/.png) from plotV1ModelMismatch. Tester testV1ModelMismatch 11/11.